Mathematical Modeling and Numerical Simulation Supplement

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1 Q1 (final: v0)

Abstract— This model materializes a reduced one-dimensional diffusion-reaction surrogate for the time-dependent actinide-line source-density proxy in expanding kilonova ejecta. It isolates delayed fission timing, expansion dilution, and free-escape boundary loss while deliberately omitting the full isotopic network, detailed atomic opacities, and multidimensional radiative transfer.

1.1 Question and Scope

Question. Can the time-dependent evolution of actinide spectral line fluxes constrain the fission neutron release timescale and distinguish kinetic decoupling from static geometric or electron-fraction abundance variations?

Reduced first executable surrogate: a one-dimensional diffusion-reaction proxy for actinide line source density in spherical ejecta. It isolates delayed fission timing and expansion dilution while not resolving the full nuclear network, detailed atomic opacities, or multidimensional radiative transfer.

1.2 Model Assumptions and Applicability

- Q1-AS-001: The evolved scalar u is a surrogate for actinide spectral-line source density rather than a full isotopic abundance network. If violated: Inferences about specific nuclear pathways or isotopic line ratios would not be supported by the surrogate.
- Q1-AS-002: Unresolved radiative transport is closed by a constant effective diffusion coefficient. If violated: Temporal spreading and escape timing of the proxy may be biased if the true transport is strongly non-diffusive.
- Q1-AS-003: Expansion is represented by a linear dilution loss with timescale formed from R_domain and v_ejecta rather than a moving mesh. If violated: The surrogate may misrepresent geometric dilution and peak-time shifts in an expanding spherical flow.
- Q1-AS-004: The initial actinide-line proxy is spatially uniform at the approved u_initial value. If violated: Initial spatial asymmetries or seed concentration gradients would not be represented.
- Q1-AS-005: The approved explicit time step is coarse relative to the approved fission-release timescale, so the delayed source is numerically smeared. If violated: The executable surrogate cannot resolve sub-step delayed-release structure and should be interpreted as a reduced kinetic proxy.

1.3 PDE Model Class

System type. DIFFUSION_REACTION_1D.
Trusted adapter. parabolic_1d.

1.4 Spatial Domain and Field Variables

Domain and mesh

- Domain: x=[0.0, 5180413674240.0]
- Grid: nx=105

Fields

- u (u); unit kg m^-3

1.5 PDE Initial and Boundary Conditions

Initial condition

- Sampled values on the normalized mesh.
- Time span: 0.0; 864000.0

Boundary condition types

- left: NEUMANN_ZERO
- right: DIRICHLET

1.6 Discretization and Stability

Discretization. FINITE_DIFFERENCE.
Time integration. EXPLICIT_EULER; time step 100.0.

1.7 Numerical Verification

The fixed PDE adapter reports finite-field checks and the family-specific stability or linear-residual diagnostic. A result is qualified only when the declared checks pass; all outcomes remain model-internal simulations.

1.8 Symbols and Units

ID	Symbol	Meaning	Unit
u	$u(x,t)$	Actinide-line source-density proxy evolved by the PDE.	kgm^-3
x	x	One-dimensional radial surrogate coordinate.	m
t	t	Time since the start of the kinetic phase.	s
D_effective	D_{eff}	Effective diffusion coefficient representing unresolved line-energy transport.	m^2s^-1
R_domain	R_{domain}	Outer radial extent of the one-dimensional surrogate domain.	m
v_ejecta	v_{ejecta}	Characteristic ejecta expansion velocity used to form the expansion dilution timescale.	ms^-1
tau_f	τ_f	Characteristic delayed fission neutron release timescale.	s
u_initial	u_{initial}	Uniform initial amplitude of the actinide-line source-density proxy.	kgm^-3
u_outer	u_{outer}	Dirichlet proxy value at the free-escape outer boundary.	kgm^-3
t_end	t_{end}	Final integration time for the executable surrogate.	s
delta_t	Δt	Temporal step size for explicit parabolic integration.	s
delta_r	Δr	Target radial grid spacing for the one-dimensional finite-difference discretization.	m

1.9 Mechanistic Equations

$$\frac{\partial u}{\partial t} = \frac{\partial}{\partial x}\left[\left(D_{\mathrm{eff}}\frac{\Delta r}{\Delta r}\right)\frac{\partial u}{\partial x}\right] + \left(\frac{u_{\mathrm{initial}}}{\Delta t}e^{-t/\tau_f}\frac{t_{\mathrm{end}}}{t_{\mathrm{end}}}\right) - \frac{v_{\mathrm{ejecta}}}{R_{\mathrm{domain}}}u \tag{1}$$

Q1-EQ-001 where u: Actinide-line source-density proxy evolved by the PDE.; t: Time since the start of the kinetic phase.; x: One-dimensional radial surrogate coordinate.; D_effective: Effective diffusion coefficient representing unresolved line-energy transport.; delta_r: Target radial grid spacing for the one-dimensional finite-difference discretization.; u_initial: Uniform initial amplitude of the actinide-line source-density proxy.; delta_t: Temporal step size for explicit parabolic integration.; tau_f: Characteristic delayed fission neutron release timescale.; t_end: Final integration time for the executable surrogate.; v_ejecta: Characteristic ejecta expansion velocity used to form the expansion dilution timescale.; R_domain: Outer radial extent of the one-dimensional surrogate domain..

$$u(x,0) = u_{\mathrm{initial}} \tag{2}$$

Q1-EQ-002 where u: Actinide-line source-density proxy evolved by the PDE.; x: One-dimensional radial surrogate coordinate.; u_initial: Uniform initial amplitude of the actinide-line source-density proxy..

$$\left.\frac{\partial u}{\partial x}\right|_{x=0} = 0 \tag{3}$$

Q1-EQ-003 where u: Actinide-line source-density proxy evolved by the PDE.; x: One-dimensional radial surrogate coordinate..

$$u(R_{\text{domain}}, t) = u_{\text{outer}} \tag{4}$$

Q1-EQ-004 where u: Actinide-line source-density proxy evolved by the PDE.; R_domain: Outer radial extent of the one-dimensional surrogate domain.; t: Time since the start of the kinetic phase.; u_outer: Dirichlet proxy value at the free-escape outer boundary..

$$S_f(t) = \frac{u_{\text{initial}}}{\Delta t} e^{-t/\tau_f} \frac{t_{\text{end}}}{t_{\text{end}}} \tag{5}$$

Q1-EQ-005 where u_initial: Uniform initial amplitude of the actinide-line source-density proxy.; delta_t: Temporal step size for explicit parabolic integration.; t: Time since the start of the kinetic phase.; tau_f: Characteristic delayed fission neutron release timescale.; t_end: Final integration time for the executable surrogate..

1.10 Initial Conditions, Boundary Conditions, and Constraints

Initial Conditions

- {'initial_condition_id': 'Q1-IC-001', 'description': 'Uniform actinide-line source-density proxy initialized to the approved u_initial amplitude.', 'profile': 'UNIFORM', 'applied_symbol': 'u_initial'}

Boundary Conditions

- {'boundary_id': 'Q1-BC-001', 'side': 'left', 'type': 'NEUMANN_ZERO', 'description': 'Zero radial flux at the symmetry center, representing the inner spherical-symmetry surrogate.'}
- {'boundary_id': 'Q1-BC-002', 'side': 'right', 'type': 'DIRICHLET', 'boundary_value': 0.0, 'description': 'Free-escape surrogate at the outer radial surface using the approved u_outer value.'}

Objective and Constraints

- {'objective_id': 'Q1-OBJ-001', 'description': 'Quantify whether temporal moments of the radial proxy separate fission-release timing from geometric or abundance perturbations.', 'target_symbol': 'u'}
- {'constraint_id': 'Q1-CON-001', 'description': 'Explicit diffusion stability margin is maintained with the diffusion Courant number below the parabolic limit.', 'threshold': 0.5}
- {'constraint_id': 'Q1-CON-002', 'description': 'The proxy density is interpreted as non-negative; negative excursions are treated as numerical artifacts.', 'threshold': 0.0}

1.11 Algorithm

Algorithm

Input

- Approved execution_ir document
- Uniform one-dimensional grid specification
- Analytic uniform initial condition
- Left and right boundary condition specification

Output

- Finite time-dependent scalar field u(x,t)
- Radial integral of u for postprocessing
- Diagnostics for stability and finiteness checks

Steps

- Construct the uniform one-dimensional spatial grid on the approved domain.
- Initialize the field with the approved uniform analytic profile.
- Evaluate the diffusion coefficient and reaction source ASTs at approved parameters.
- Advance the diffusion-reaction system with the registered explicit parabolic integrator.
- Enforce zero-flux symmetry at the left boundary and free-escape Dirichlet condition at the right boundary.
- Store the evolved field and compute radial summary quantities for later comparison.

1.12 Parameters, Scenarios, and Numerical Settings

Parameters

- {'parameter_id': 'domain_radius', 'mathir_symbol': 'R_domain', 'approved_value': 5180413674240.0, 'unit': 'm', 'role': 'SCENARIO_INPUT', 'provenance': 'APPROVED_LITERATURE_SINGLE_SOURCE'}
- {'parameter_id': 'ejecta_velocity', 'mathir_symbol': 'v_ejecta', 'approved_value': 59958491.6, 'unit': 'm s^-1', 'role': 'SCENARIO_INPUT', 'provenance': 'APPROVED_LITERATURE_SINGLE_SOURCE'}
- {'parameter_id': 'fission_neutron_release_timescale', 'mathir_symbol': 'tau_f', 'approved_value': 2.3, 'unit': 's', 'role': 'MATERIAL_PROPERTY', 'provenance': 'APPROVED_LITERATURE_SINGLE_SOURCE'}
- {'parameter_id': 'initial_actinide_density_proxy', 'mathir_symbol': 'u_initial', 'approved_value': 5.12186785866739e-14, 'unit': 'kg m^-3', 'role': 'SCENARIO_INPUT', 'provenance': 'APPROVED_LITERATURE_SINGLE_SOURCE'}

- {'parameter_id': 'effective_diffusion_coefficient', 'mathir_symbol': 'D_effective', 'approved_value': 1e+18, 'unit': 'm^2 s^-1', 'role': 'MATERIAL_PROPERTY', 'provenance': 'APPROVED_MODEL_ASSUMPTION'}
- {'parameter_id': 'outer_boundary_proxy_value', 'mathir_symbol': 'u_outer', 'approved_value': 0.0, 'unit': 'kg m^-3', 'role': 'BOUNDARY_CONDITION', 'provenance': 'APPROVED_MODEL_ASSUMPTION'}
- {'parameter_id': 'simulation_end_time', 'mathir_symbol': 't_end', 'approved_value': 864000.0, 'unit': 's', 'role': 'MODEL_ASSUMPTION', 'provenance': 'APPROVED_MODEL_ASSUMPTION'}
- {'parameter_id': 'time_step_size', 'mathir_symbol': 'delta_t', 'approved_value': 100.0, 'unit': 's', 'role': 'MODEL_ASSUMPTION', 'provenance': 'APPROVED_MODEL_ASSUMPTION'}
- {'parameter_id': 'radial_grid_spacing', 'mathir_symbol': 'delta_r', 'approved_value': 50000000000.0, 'unit': 'm', 'role': 'MODEL_ASSUMPTION', 'provenance': 'APPROVED_MODEL_ASSUMPTION'}

Scenarios

- {'scenario_id': 'Q1-SC-001', 'name': 'Instantaneous fission recycling', 'description': 'Conceptual limit in which the fission source is deposited essentially within the first numerical step.', 'mathematical_difference': 'Early-time radial integral over the first step is compared against later-time integrals.', 'execution_status': 'CONCEPTUAL_LIMIT'}
- {'scenario_id': 'Q1-SC-002', 'name': 'Delayed fission recycling', 'description': 'Baseline executable surrogate uses the approved tau_f in the exponential source factor.', 'mathematical_difference': 'First temporal moment of the radial integral is used to quantify delay.', 'execution_status': 'BASELINE'}
- {'scenario_id': 'Q1-SC-003', 'name': 'Geometric asymmetry only', 'description': 'Conceptual geometric comparison represented by radial variance of the proxy in postprocessing.', 'mathematical_difference': 'Radial variance replaces temporal moment as the discriminator.', 'execution_status': 'CONCEPTUAL_POSTPROCESSING'}
- {'scenario_id': 'Q1-SC-004', 'name': 'Variable Ye only', 'description': 'Conceptual abundance comparison represented by the domain-integrated proxy mass.', 'mathematical_difference': 'Domain integral of u is used as the abundance discriminator.', 'execution_status': 'CONCEPTUAL_POSTPROCESSING'}

Solver family. parabolic_1d.

1.13 Parameter Provenance and Applicability

Approved parameter-set identity. 8dd7a4bea03faa204d37b8fd38bf995e22672f0964d57af785767954ed827f3e.

- domain_radius (R_domain) = 5180413674240.0 m; role SCENARIO_INPUT; status APPROVED_LITERATURE_SINGLE_SOURCE; source: Radioactivity and

Thermalization in the Ejecta of Compact Object Mergers and Their Impact on Kilonova Light Curves; DOI 10.3847/0004-637x/829/2/110; document PFD-002; locator: preprint; II.1 Ejecta Model; page 3; applicability: composition=neutron-rich r-process ejecta; epoch_days=1.0; geometry=spherical homologous ejecta; required_condition_1=geometry: spherical radial surrogate; required_condition_2=composition: actinide-bearing kilonova ejecta; required_condition_3=epoch: early post-merger kinetic phase. uncertainty: reference_epoch=1 day; velocity_range=0.1-0.3 c.

- ejecta_velocity (v_ejecta) = 59958491.6 m_s^-1; role SCENARIO_INPUT; status APPROVED_LITERATURE_SINGLE_SOURCE; source: Opacities and Spectra of the r-Process Ejecta from Neutron Star Mergers; DOI 10.1088/0004-637x/774/1/25; document PFD-014; locator: preprint; I. Introduction; page 1; applicability: composition=neutron-rich r-process ejecta; expansion=homologous; required_condition_1=expansion: homologous ejecta surrogate; required_condition_2=composition: actinide-bearing kilonova ejecta; velocity_range=0.1-0.3 c. uncertainty: reported_range=0.1-0.3 c.

- fission_neutron_release_timescale (tau_f) = 2.3 s; role MATERIAL_PROPERTY; status APPROVED_LITERATURE_SINGLE_SOURCE; source: β-delayed Fission in r-process Nucleosynthesis; DOI 10.3847/1538-4357/aaeaca; document PFD-018; locator: preprint; III. Results; Figure 4; page 5; applicability: kinetic_regime=late fission-recycling phase; nuclear_regime=beta-delayed fission in neutron-rich r-process nuclei; required_condition_1=nuclear regime: actinide fission recycling; required_condition_2=kinetic regime: delayed neutron release relative to expansion. uncertainty: interpretation=abundance-weighted transition epoch used as a characteristic proxy, not an isotope-specific half-life.

- initial_actinide_density_proxy (u_initial) = 5.12186785866739e-14 kg_m^-3; role SCENARIO_INPUT; status APPROVED_LITERATURE_SINGLE_SOURCE; source: NLTE Spectra of Kilonovae; DOI 10.1093/mnras/stad3106; document PFD-026; locator: article; 5. Discussion; Table A1; page 19; applicability: composition=Th and U actinides in a homogeneous kilonova model; electron_fraction=Ye approximately 0.15; normalization_epoch_days=1.0; required_condition_1=initial state: seed nuclei distribution at the start of the kinetic phase; required_condition_2=composition: actinide-bearing ejecta component. uncertainty: actinide_fraction=0.003 combined Th+U fraction at Ye approximately 0.15; density_model=uniform spherical proxy.

- effective_diffusion_coefficient (D_effective) = 1e+18 m^2_s^-1; role MATERIAL_PROPERTY; status APPROVED_MODEL_ASSUMPTION; source: MODEL ASSUMPTION (not a measured or literature value); applicability: required_condition_1=transport approximation: diffusion surrogate for line-energy propagation; required_condition_2=closure assumption: temperature and opacity dependence collapsed into an effective constant. uncertainty: not reported.

- outer_boundary_proxy_value (u_outer) = 0.0 kg_m^-3; role BOUNDARY_CONDITION; status APPROVED_MODEL_ASSUMPTION; source: MODEL ASSUMPTION (not a measured or literature value); applicability: required_condition_1=boundary geometry: outer radial surface; required_condition_2=boundary physics: free-streaming or escape surrogate. uncertainty: not reported.

- simulation_end_time (t_end) = 864000.0 s; role MODEL_ASSUMPTION; status APPROVED_MODEL_ASSUMPTION; source: MODEL ASSUMPTION (not a measured or literature value); applicability: required_condition_1=numerical control: finite simulation window; required_condition_2=scientific window: covers delayed fission kinetic signature. uncertainty: not reported.

- time_step_size (delta_t) = 100.0 s; role MODEL_ASSUMPTION; status APPROVED_MODEL_ASSUMPTION; source: MODEL ASSUMPTION (not a measured or literature value); applicability: required_condition_1=numerical control: explicit time integration; required_condition_2=stability requirement: parabolic diffusion surrogate. uncertainty: not reported.

- radial_grid_spacing (delta_r) = 50000000000.0 m; role MODEL_ASSUMPTION; status APPROVED_MODEL_ASSUMPTION; source: MODEL ASSUMPTION (not a measured or literature value); applicability: required_condition_1=numerical control: finite-difference radial grid; required_condition_2=geometry: one-dimensional spherical radial surrogate. uncertainty: not reported.

1.14 Numerical Validation and Model-Internal Results

Validation Plan

- {'validation_id': 'Q1-VAL-001', 'description': 'Check that the evolved field remains finite over the full approved time span.', 'expected_behavior': 'No non-finite values appear in the field or expressions.'}

- {'validation_id': 'Q1-VAL-002', 'description': 'Check that the zero-flux inner boundary preserves symmetry in the radial surrogate.', 'expected_behavior': 'No artificial flux enters through the left boundary.'}

- {'validation_id': 'Q1-VAL-003', 'description': 'Check that the outer Dirichlet boundary permits proxy escape toward the approved free-escape value.', 'expected_behavior': 'The field near the outer boundary relaxes toward the approved outer value.'}

- {'validation_id': 'Q1-VAL-004', 'description': 'Check that the explicit diffusion step size satisfies the approved stability margin.', 'expected_behavior': 'The diffusion Courant number remains safely below the parabolic limit.'}

- Execution sim-48b8db888a0947a4a806d38e69529075; NUMERICAL_SIMULATION; SIMULATED; NOT_EMPIRICAL; relation INCONCLUSIVE; summary: The baseline one-dimensional diffusion-reaction simulation completed and was numerically verified. The final field remained near 10^-18, with finite-field, explicit-stability, and zero-boundary-residual checks passing. This is a model-internal numerical result and does not establish observational confirmation.

1.15 Hypothesis Iteration Lineage

- v0: INCONCLUSIVE (The baseline one-dimensional diffusion-reaction simulation completed and was numerically verified. The final field remained near 10^-18, with finite-field, explicit-stability, and zero-boundary-residual checks passing. This is a model-internal numerical result and does not establish observational confirmation.)

1.16 Limitations, Non-Empirical Disclosure, and References

All values in this section are model-internal numerical simulations (NUMERICAL_SIMULATION; SIMULATED; NOT_EMPIRICAL), not empirical observations.

Limitations

- The surrogate does not resolve a full r-process isotopic network or actinide-specific line formation.
- The effective diffusion coefficient is a declared closure assumption, not a literature measurement.
- The coarse approved time step cannot resolve all structure on the approved fission-release timescale.
- Spherical symmetry is assumed in the reduced radial surrogate and cannot represent multidimensional lanthanide curtaining.
- Actinide opacity and atomic-data uncertainties are not explicitly represented.

References

- {’reference_id’: ’Q1-REF-001’, ’citation’: ’Barnes et al., Radioactivity and Thermalization in the Ejecta of Compact Object Mergers and Their Impact on Kilonova Light Curves, 2016’, ’relevance’: ’Supports the homologous-expansion radius proxy used for the domain radius.’}
- {’reference_id’: ’Q1-REF-002’, ’citation’: ’Kasen et al., Opacities and Spectra of the r-Process Ejecta from Neutron Star Mergers, 2013’, ’relevance’: ’Supports the characteristic ejecta velocity range used for expansion dilution.’}
- {’reference_id’: ’Q1-REF-003’, ’citation’: ’Mumpower et al., beta-delayed Fission in r-process Nucleosynthesis, 2018’, ’relevance’: ’Provides the abundance-weighted kinetic proxy for delayed fission recycling.’}
- {’reference_id’: ’Q1-REF-004’, ’citation’: ’Domato et al., NLTE Spectra of Kilonovae, 2023’, ’relevance’: ’Supports the actinide mass-fraction normalization used for the initial density proxy.’}

2 Q2 (final: v0)

Abstract— An executable scalar Monte Carlo posterior-proxy model for the fission neutron release timescale tau_f. The run samples tau_f together with ejecta mass, electron fraction, and inclination nuisance priors using the approved Q2 parameter values. The current trusted adapter reports the sampled tau_f observable and does not claim a full hierarchical likelihood, Markov-chain diagnostic, or empirical posterior.

2.1 Question and Scope

Question. What is the posterior probability distribution of the fission neutron release timescale when gravitational-wave, kilonova photometric, and spectroscopic abundance constraints are combined?

A bounded zero-dimensional Monte Carlo posterior-proxy calculation for one kilonova event. The full multi-event hierarchical likelihood, parallel tempering, Bayes-factor estimation, and catalogue-level population aggregation remain outside the current trusted solver.

2.2 Model Assumptions and Applicability

- Q2-A-001: The approved tau_f value of 2.3 s is used as the center of a positive bounded sensitivity prior from 1.15 s to 3.45 s. If violated: The sampled fission-timescale sensitivity range changes.
- Q2-A-002: The approved ejecta-mass value of 0.04 solar_mass is represented by a positive bounded interval from 0.03 to 0.05 solar_mass. If violated: The nuisance ejecta-mass prior changes.
- Q2-A-003: The approved electron-fraction value of 0.16 is represented by a broad physically admissible interval from 0.08 to 0.24. If violated: The neutron-richness nuisance prior changes.
- Q2-A-004: The approved inclination value is represented by the converted 19 to 42 degree interval in radians. If violated: The viewing-angle nuisance prior changes.
- Q2-A-005: The current executable observable is the sampled tau_f value; nuisance variables are retained as declared prior variables for later likelihood extension. If violated: The result would no longer be the scalar tau_f posterior proxy defined by this model.

2.3 Symbols and Units

ID	Symbol	Meaning	Unit
Q2-S-001	τ_f	Fission neutron release timescale	s
Q2-S-002	M_{ej}	Kilonova ejecta mass nuisance parameter	solar mass
Q2-S-003	Y_e	Electron fraction nuisance parameter	1
Q2-S-004	i	Binary inclination angle	rad
Q2-S-005	N_{samples}	Monte Carlo sample count	1
Q2-S-006	N_{chains}	Declared independent-chain control	1
Q2-S-007	$N_{\text{temperatures}}$	Declared tempering-level control	1

2.4 Mechanistic Equations

$$z = \tau_f \tag{6}$$

Q2-EQ-001 where Q2-S-001: Fission neutron release timescale.

$$p(\tau_f, M_{\text{ej}}, Y_e, i) = p(\tau_f)p(M_{\text{ej}})p(Y_e)p(i) \tag{7}$$

Q2-EQ-002 where Q2-S-001: Fission neutron release timescale; Q2-S-002: Kilonova ejecta mass nuisance parameter; Q2-S-003: Electron fraction nuisance parameter; Q2-S-004: Binary inclination angle.

2.5 Initial Conditions, Boundary Conditions, and Constraints

Initial Conditions

- The zero-dimensional sampler is initialized by drawing each declared random variable from its validated prior distribution.

Boundary Conditions

- No spatial boundary conditions apply to this zero-dimensional Monte Carlo sampler.

Objective and Constraints

- Estimate the scalar sampled distribution of tau_f under the declared bounded nuisance-prior setup.

2.6 Algorithm

Algorithm

Input

- tau_f prior
- M_ej prior
- Y_e prior
- i prior
- approved parameter set

Output

- sample count
- mean tau_f
- standard deviation of tau_f
- minimum tau_f
- maximum tau_f

Steps

- Initialize the deterministic random generator from the integer seed.
- Draw 10000 independent samples from the four declared bounded prior distributions.
- Evaluate the scalar observable z=tau_f for every draw.
- Report bounded summary statistics for the sampled tau_f values.

2.7 Parameters, Scenarios, and Numerical Settings

Parameters

- The execution parameter object must contain exactly tau_f=2.3, M_ej=0.04, Y_e=0.16, i=0.5585053606381855, N_samples=10000, N_chains=4, and N_temperatures=8.
- The random-variable sensitivity bounds are explicit model assumptions and are numeric literals required by the trusted sampler.

Scenarios

- baseline

Solver family. MONTE_CARLO.

2.8 Parameter Provenance and Applicability

Approved parameter-set identity. a5e0912a88039b2cfdeebd2803aee0f02f55493c372243df d3a461294867eb1e.

- tau_f (tau_f) = 2.3 s; role MATERIAL_PROPERTY; status APPROVED_ LITERATURE_SINGLE_SOURCE; source: beta-delayed Fission in r-process Nucleosynthesis; DOI 10.3847/1538-4357/aaeaca; document SEARCH-Q2-TAU-F-2018; locator: ar5iv_fulltext; Results and discussion; Figure 4 discussion; applicability: positive_ timescale=True; posterior_parameter=abundance-weighted kinetic proxy; required_ condition_1=posterior_parameter; required_condition_2=positive_timescale. uncertainty: reported=approximately 2.3 s; no formal interval in the quoted passage.
- M_ej (M_ej) = 0.04 solar_mass; role SCENARIO_INPUT; status APPROVED_ LITERATURE_SINGLE_SOURCE; source: A kilonova as the electromagnetic counterpart to a gravitational-wave source; DOI 10.1038/nature24303; document SEARCH-Q2-M-EJ-2017; locator: controlled_fulltext_document; Ejecta mass estimate; applicability: positive_mass=True; posterior_parameter=event ejecta-mass estimate used as a nuisance-parameter center; required_condition_1=posterior_parameter; required_ condition_2=positive_mass. uncertainty: lower=0.01; unit=solar_mass.
- Y_e (Y_e) = 0.16 1; role SCENARIO_INPUT; status APPROVED_LITERATURE_ SINGLE_SOURCE; source: Long-term GRMHD simulations of neutron star merger accretion discs: implications for electromagnetic counterparts; DOI 10.1093/mnras/sty2932; document SEARCH-Q2-YE-2018; locator: openalex_anysearch_abstract; Abstract; applicability: physically_admissible_electron_fraction=True; posterior_ parameter=representative literature prior center; required_condition_1=posterior_ parameter; required_condition_2=physically_admissible_electron_fraction. uncertainty: reported=broad distribution; quoted value is the lower average, not a narrow posterior interval.
- i (i) = 0.5585053606381855 rad; role SCENARIO_INPUT; status APPROVED_ LITERATURE_SINGLE_SOURCE; source: Measuring the Viewing Angle of GW170817 with Electromagnetic and Gravitational Waves; DOI 10.3847/2041-8213/aac6c1; document SEARCH-Q2-INCLINATION-2018; locator: ar5iv_fulltext; III Results; applicability:

geometric_admissible_inclination=True; posterior_parameter=joint GW-EM Bayesian viewing-angle posterior; required_condition_1=posterior_parameter; required_condition_2=geometric_admissible_inclination. uncertainty: lower_degrees=13.0; median_degrees=32.0; systematic_degrees=1.7; upper_degrees=10.0.

- n_mcmc_samples (N_samples) = 10000.0 1; role MODEL_ASSUMPTION; status APPROVED_MODEL_ASSUMPTION; source: MODEL ASSUMPTION (not a measured or literature value); applicability: required_condition_1=solver_control. uncertainty: not reported.
- n_chains (N_chains) = 4.0 1; role MODEL_ASSUMPTION; status APPROVED_MODEL_ASSUMPTION; source: MODEL ASSUMPTION (not a measured or literature value); applicability: required_condition_1=solver_control. uncertainty: not reported.
- n_tempering_levels (N_temperatures) = 8.0 1; role MODEL_ASSUMPTION; status APPROVED_MODEL_ASSUMPTION; source: MODEL ASSUMPTION (not a measured or literature value); applicability: required_condition_1=solver_control. uncertainty: not reported.

2.9 Numerical Validation and Model-Internal Results

Validation Plan

- Verify that the sample count is an integer between 1 and 100000.
- Verify that every distribution bound is finite and ordered.
- Verify that all observable values are finite.
- Verify that the execution parameter object exactly matches the approved parameter set.
- Execution sim-1242bc317d55543009aa59d9c4632170f; NUMERICAL_SIMULATION; SIMULATED; NOT_EMPIRICAL; relation INCONCLUSIVE; summary: Q2

2.10 Hypothesis Iteration Lineage

- v0: INCONCLUSIVE (Q2)

2.11 Limitations, Non-Empirical Disclosure, and References

All values in this section are model-internal numerical simulations (NUMERICAL_SIMULATION; SIMULATED; NOT_EMPIRICAL), not empirical observations.

Limitations

- The current trusted adapter evaluates a scalar observable and does not implement a full likelihood.
- The declared N_chains and N_temperatures controls are retained for provenance but are not independently executed by the standard adapter.
- The output is a numerical simulation result and not empirical evidence.
- The nuisance priors are sampled but are not combined through a multi-messenger likelihood in this adapter.

References

- doi:10.3847/1538-4357/aaeaca
- doi:10.1038/nature24303
- doi:10.1093/mnras/sty2932
- doi:10.3847/2041-8213/aac6c1